

JAPNIT SINGH SAWHNEY

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EDUCATION

Thapar Institute Of Engineering & Technology, Patiala	Aug 2023 - Present
B.Tech in Computer Engineering	CGPA:9.10
St.Mary's Academy, Meerut	Apr 2022 - Mar 2023
Senior Secondary School certificate	Percentage:94.5%
St.Mary's Academy, Meerut	Apr 2020 - Mar 2021
Senior Secondary School certificate	Percentage:95.8%

PROJECTS

AI Interview Coach



- Built a modular AI interview platform using LangGraph-based multi-agent workflows to generate personalized interview questions, evaluate responses, and produce structured interview reports from resume and job-description context.
- Designed a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB, HuggingFace embeddings, and Groq LLMs with speech interfaces and modular agents for question generation, evaluation, and reporting.
- **Tech Stack** -Python, LangChain, FastAPI, HuggingFace Embeddings, RAG, Speech-to-Text, Text-to-Speech

UniVerse - CampusGPT Assistant for TIET Students



- Built a hybrid AI assistant integrating LLM-based reasoning with structured MySQL services and JSON knowledge sources to answer real-time campus queries through intelligent intent routing.
- Designed graph-based campus navigation using weighted graphs, Dijkstra's algorithm, OSRM routing, and Leaflet maps with real-time walking time estimation.
- **Tech Stack** - MySQL, Google Gemini API, Vercel, Render, Python

Bangalore Traffic Prediction



- Built a Bangalore traffic congestion prediction system using historical traffic, weather, environmental impact, and temporal features, deployed via a FastAPI backend to deliver real-time congestion levels, categories, and analytical insights.
- Implemented a stacked Random Forest–XGBoost regression model with spatial and temporal feature engineering, achieving $RMSE = 2.031$ and $R^2 = 0.995$, outperforming baseline models and results reported in related research studies.
- **Tech Stack** - Python, FastAPI, Google Gemini API, HTML, CSS

TECHNICAL SKILLS

Languages: Python, Java, C++, C

Backend & API: FastAPI, REST APIs

Artificial Intelligence/Machine Learning: LangChain, LangGraph, Agentic AI, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, ChromaDB, Scikit-learn, NumPy, Pandas

Databases: MySQL, ChromaDB

Tools & Technologies: Git, GitHub, Linux, Ollama, Render, Vercel, VS Code

Core CS Concepts: Object-Oriented Programming (OOP), Operating Systems (OS), DBMS, Data Structures and Algorithms, Computer Networks

ACHIEVEMENTS

Selected for ML Summer School 2026

Solved 350+ Data Structures & Algorithms problems on LeetCode.

Gold medalist in 3600m cycle race.

CERTIFICATION

EC-Council Cybersecurity Workshop (1 CPE Credit)